My Database Theory

- Data vs Information

Data refers to raw, unprocessed facts, figures, or symbols that, on their own, may not have much meaning. For example, the number “42” is a piece of data. Information, on the other hand, is data that has been processed, organised, and given context to provide meaning.

- What is metadata?

Metadata is data that describes data (ie. It helps with interpretation and meaning making)

- What is a DBMS and what are its advantages?

A Database Management System (DBMS) is specialized software that serves as an interface between a database and its end-users or applications. It allows users to create, retrieve, update, and manage data in a structured, efficient, and secure manner. Instead of writing raw code to manipulate data directly on a hard drive, a DBMS provides standardized tools (such as SQL—Structured Query Language) to interact with the data

- Compare and contrast operational databases with analytical databases, and provide an example of each

Operational databases are designed to support the day-to-day operations of an organisation. They store and manage current, detailed data essential for running the business. Often referred to as **OLTP (online transaction processing)** databases, they are optimised for speed and efficiency in handling frequent updates and retrievals. Examples of where an operational database may be used are inventory management, order processing, and financial transactions. Analytical databases, often referred to as **OLAP (online analytical processing)** databases, are designed for data analysis and business intelligence. Unlike operational databases, they store historical and aggregated data. These databases are optimised for complex queries and reporting, allowing businesses to gain insights into trends, patterns, and performance. Examples of analytical databases include data warehouses, data marts, and online analytical cubes, which support decision-making processes across various departments within an organisation.

- Explain the types of data and use cases where NoSQL databases are most effective

Non-relational databases, often referred to as NoSQL (Not only SQL) databases, offer a more flexible approach to data storage and management. Unlike their relational counterparts, NoSQL databases don't enforce a rigid structure, allowing for diverse data

models and scalability. They are designed to handle massive amounts of unstructured or semi-structured data.

- Which DBMS does not require server configuration, and what are the advantages of using it?

SQLite is a lightweight, fi le-based relational database that does not require a server configuration. Its simplicity and self-contained nature make it ideal for mobile applications, embedded systems, and small-scale projects.

- Explain ACID properties in the context of a DBMS

ACID stands for atomicity, consistency, isolation, and durability.

● Atomicity is when data can not be divided into smaller units.

● Consistency refers to the correctness of the database.

● Isolation means that parallel transactions should be able to happen without affecting each other.

● Durability implies that changes are permanent and can be recovered in the event of a system crash.

Essentially, ACID guarantees that database transactions are complete, accurate, independent, and permanent, even in the event of system failures. While some NoSQL databases offer ACID compliance for specific use cases, they are generally known for prioritising other characteristics like scalability and performance.